

Fares Elsobky

Airbus Defense & Space
Software Engineer

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SKILLS

Languages: Python | Java SQL | JavaScript | C++ | SQL

Backend / Web: Spring Boot | Spring Cloud | Angular | REST APIs | OAuth 2.0

AI / Data: PyTorch | TensorFlow | Deep Learning | Reinforcement Learning | Data Analysis | Bayesian Optimization

DevOps / Cloud: Docker | Docker Compose | Kubernetes | GitLab CI/CD

Testing / Tools: Playwright | Rest Assured | Git | Jira

Other: SAP ABAP | SAPUI5 | OData | Raspberry Pi | PySceneDetect

EDUCATION & TRAINING

Technical University of Munich (TUM)

Munich/Germany

Master Of Science: Data Engineer and Analytics

Graduation 2026

- **Major: Data Engineering and Analytics , Minor: Software Engineering**
- **Master's Thesis: AI-Based Access Point Placement Optimization in Indoor Wireless Networks**
 - Developed a simulation-based optimization framework for indoor Access Point placement using Reinforcement Learning and Bayesian Optimization.
 - Built automated Python pipelines for experiments, benchmarking, and result analysis.
 - Evaluated performance using runtime, coverage, and feasibility metrics across multiple scenarios.
 - Improved optimization speed using TD3-based RL compared to traditional methods.
 - Applied machine learning, data analysis, and software engineering in a research environment.

Arab Academy for Science, Technology & Maritime Transport (AASTMT)

Cairo/Egypt

Bachelor's in Computing and Information Technology 3.4 (US GPA)

Graduation 2021

- **Major: Computer Science (CS) - Software Engineering (SE)**
- **Bachelor Thesis (A+): REAL-TIME CAR VIOLATIONS DETECTION SYSTEM**
 - Developed a real-time car violation detector using Raspberry Pi with cameras, IR, GPS, AI, OpenCV, and cloud computing.
 - Detected traffic light, speed, and lane violations using OpenCV and presented them on Azure automatically.
 - Used Google Maps API for location and speed, and built a Python Flask website connected to an Azure database for user access.
 - Reduced road crash risks through advanced violation detection.

PROFESSIONAL EXPERIENCE & PROJECTS

Airbus Defense & Space

Munich/Germany

Software Engineer: Nomination Tool Developer

Oct 2022 – Oct 2024

- **Developed** and maintained an in-house Nomination Tool using Google App Script (JavaScript), achieving end-to-end digitization of nomination processes and significantly enhancing efficiency.
- **Tested** each version before release to ensure reliability and performance, and **handled** user problems and errors to maintain a smooth user experience.
- **Deployed** new releases and **upgraded** the tool from Version 6 to Version 11, continuously adding new features based on stakeholder requirements to improve functionality and user experience.
- **Enhanced** the tool by integrating a new validation layer for nominations, **automated** data collection from Google Workspace, and **streamlined** the creation of forms and emails to reduce manual overhead.
- **Improved** system user experience through tool modulation and extensive automation, ensuring data coherence and supporting the team in their operational activities by effectively interacting with key stakeholders.

- **Managed** data coherency for the department, ensured accurate and timely data handling, and **provided** support to the team in their daily operations.
- **Drove** system improvements by continually assessing and optimizing the tool's performance, incorporating user feedback to enhance overall efficiency and satisfaction.

TUM & CapGemini

Advanced Practical Course - Enterprise Software Engineering using SAP

Munich/Germany
Oct 2023 – Feb 2024

- **Implemented** a cockpit project that aimed to create a robust sustainability application using ABAP for backend functionality, OData for seamless data exchange, and SAPUI5 for frontend visualization.
- **Met** with stakeholders to collect project requirements and translate them into a product.
- **Created** an Agile roadmap, data model, and software architecture.
- **Held** weekly meetings with stakeholders to present progress and refine the roadmap accordingly.
- The cockpit project **included** a Company Visualization Dashboard, an intuitive high-level dashboard, and functionalities to trace recent data and analyze goal achievement.

TUM - Data Processing

High-Level Feature Extraction Application

Munich/Germany
Nov 2024 – May 20

- **Developing** an application to extract high-level features from images using advanced machine learning models.
- **Model Utilization:** Leveraging BLIP-2 for image captioning and integrating models from Hugging Face and Model Zoo, with plans to incorporate BLIP-3 for video captioning, along with fine-tuning enhancements.
- **Scene Detection:** Implementing PySceneDetect to identify and select keyframes from video scenes.
- **Feature Aggregation:** Combining keyframes into composite images for comprehensive feature extraction.
- **Data Management:** Storing extracted high-level features and integrating the feature extraction module with an existing tool to enhance functionality.

TUM - Advanced Topics in Software Engineering Course

Online IDE Development

Munich/Germany
Oct 2024 – April 2025

Collaborating on the development of an online Integrated Development Environment (IDE) as part of an advanced software engineering curriculum.

- **CI/CD:** Utilized Docker for virtualization, Docker Compose for container orchestration, and GitLab CI/CD for continuous integration pipelines.
- **Frontend (Angular):** Designed component architecture and user interfaces; applied the Observer Pattern for distributed event handling.
- **Microservices (Spring Cloud):** Implemented service discovery and API gateway patterns; ensured load balancing and adopted a container-per-service approach.
- **Backend (Spring Boot):** Develop RESTful web services, Spring Security, managed data access with JDBC/JPA/Hibernate, implemented OAuth 2.0 authentication, and conducted unit/integration testing.

ADDITIONAL

LANGUAGES: Arabic (native), English (fluent)